

Project Design Phase-II

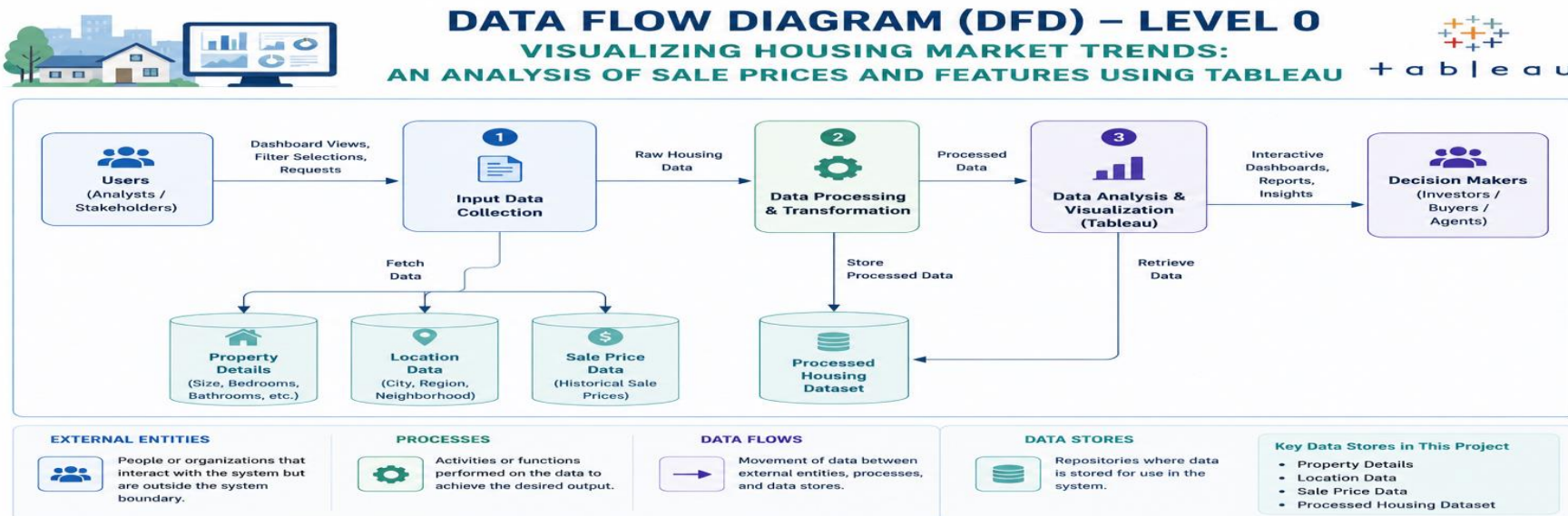
Data Flow Diagram & User Stories

Date	31 January 2025
Team ID	
Project Name	visualizing housing market trends:an analysis of sale prices and features using tableau
Maximum Marks	4 Marks

Data Flow Diagrams:

A **Data Flow Diagram (DFD)** is a graphical representation that illustrates how data moves through the housing market analysis system. For the project "**Visualizing Housing Market Trends: An Analysis of Sale Prices and Features Using Tableau**," the DFD represents the flow of housing data from data collection and preprocessing to analysis, visualization, and reporting. It clearly shows how property details and sale price information.

Example: [\(Simplified\)](#)



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Analyst / User	Dashboard Access	USN-1	As a user, I can open the housing market dashboard to analyze property data.	User can successfully access the dashboard.	High	Sprint-1
Analyst / User	Data Visualization	USN-2	As a user, I can view interactive charts showing house sale prices by different features.	Charts load correctly and display accurate information.	High	Sprint-1
Analyst / User	Filter Data	USN-3	As a user, I can filter housing data by bedrooms, bathrooms, floors, and renovation status.	Filters update all dashboard visualizations correctly.	High	Sprint-1
Analyst / User	Trend Analysis	USN-4	As a user, I can analyze housing market trends over different years.	Trend charts display accurate yearly comparisons.	High	Sprint-1
Analyst / User	Price Analysis	USN-5	As a user, I can compare sale prices based on house features.	Dashboard displays feature-wise price comparisons	High	Sprint-1
Analyst / User	Geographic Analysis	USN-6	As a user, I can view house prices based on location.	Map visualization displays location-wise sale prices.	Medium	Sprint-2
Analyst / User	Dashboard Interaction	USN-7	As a user, I can interact with charts by selecting values and viewing details.	Tooltips and interactive filtering work correctly.	Medium	Sprint-2

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Analyst / User	Data Insights	USN-8	As a user, I can identify the factors that most influence housing prices.	Dashboard provides meaningful visual insights.	High	Sprint-2
Analyst / User	Report Generation	USN-9	As a user, I can export dashboard views for reporting purposes.	Dashboard can be exported successfully.	Medium	Sprint-3
Administrator	Dataset Management	USN-10	As an administrator, I can update the housing dataset used in the dashboard.	Updated data is reflected accurately in all visualizations.	High	Sprint-3

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

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Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Collection	Import housing sales dataset (CSV/Excel) Load property features such as price, location, area, bedrooms, bathrooms, year built
FR-2	Data Preparation	Clean missing values Remove duplicate records Transform and format data for Tableau analysis
FR-3	Data Visualization	Create interactive dashboards Display charts for sale prices, property features, location-wise trends, and market comparisons
FR-4	Trend Analysis	Analyze relationships between sale prices and housing features Identify market trends using filters and calculated fields
FR-5	Dashboard Interaction	Provide filters for city, property type, year, and price range Enable drill-down and interactive exploration
FR-6	Reporting & Insights	Generate summary reports Highlight key findings and support data-driven decision making

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Easy-to-use and interactive Tableau dashboards.
NFR-2	Security	Protect housing data from unauthorized access.
NFR-3	Reliability	Ensure accurate and consistent data visualization.
NFR-4	Performance	Load dashboards and charts quickly.
NFR-5	Availability	Make dashboards accessible whenever needed
NFR-6	Scalability	Support larger datasets without performance issues

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	31 January 3035
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Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>

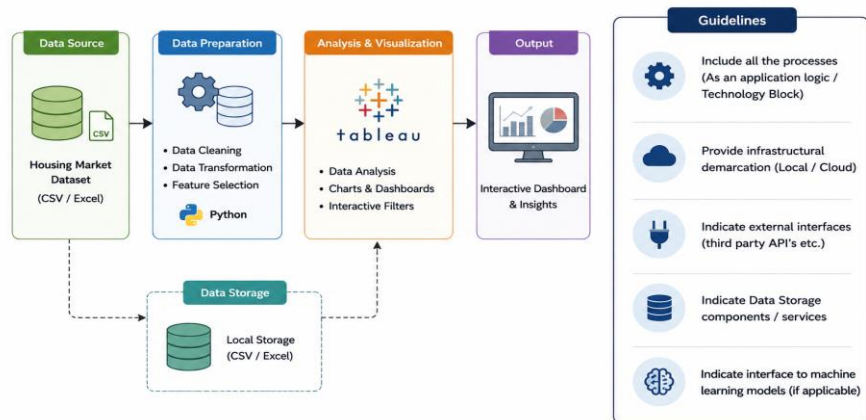


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Dashboard for viewing housing market insights	Tableau
2.	Data Processing	Clean and prepare housing dataset	Microsoft Excel / Python (Optional)
3.	Data Analysis	Analyze sale prices and housing features	Tableau Calculated Fields
4.	Data Visualization	Create interactive charts and dashboards	Tableau Desktop
5.	Data Source	Store housing sales dataset	CSV / Excel
6.	Data Connection	Connect Tableau with dataset	Tableau Data Connector
7.	File Storage	Store project datasets	Local File System
8.	External Data	Housing market dataset	Kaggle Housing Dataset
9.	Data Filters	Filter by location, price, bedrooms, etc	Tableau Filters

10.	Analytics	Trend and comparative analysis	Tableau Analytics
11.	Deployment	Share dashboards and reports	Tableau Public / Tableau Desktop

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Framework	Data visualization platform	Tableau Public
2.	Security	Secure access to project files	Password Protection
3.	Scalable Architecture	Supports large housing datasets	Tableau Desktop
4.	Availability	Accessible on desktop and web	Tableau Public
5.	Performance	Fast dashboard loading and filtering	Tableau Analytics

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>

